Features left to build: NHA M1 assessment

HMS ABHA integration (ABHA creation and verification). Based on the NHA test-case sheet, as of 18 Oct 2026, assuming we are a **private integrator**. Effort is a rough guide: **Easy** = hours to a day, **Medium** = a few days, **Hard** = a week or more, or blocked on something outside our code.

Group	Count	In short
Mandatory, not done	3	Scan and Share; ABHA-address login by Aadhaar OTP and by mobile OTP
Mandatory, built but not tested with a real OTP	14	Needs testing, then fixes for whatever the real OTPs show
Mandatory, small gaps inside rows we marked as built	3	Full profile display, 'address already exists' message, OTP wording
Optional, easy	5	Multilingual consent, mobile/email update, ABDM logout, and more
Optional, medium	8	Password login, photo update, re-KYC, delete ABHA, QR scan, tests, and more
Optional, hard	4	Driving licence / PAN, biometrics, child ABHA, PHR address creation

1. Mandatory and still failing (3)

Sheet row	Feature	Effort	What it needs
SHARE_PATIENT_PROFILE_701	**Scan and Share.** The patient scans a QR at the facility, the profile reaches our system, and the patient gets a token number valid for 30 minutes.	Hard	A publicly reachable HTTPS backend (so deploy first). Register our bridge URL with ABDM. Receiving endpoints patient-share/share and on-share. A facility QR built from our HFR ID. A token-number screen at the desk.
VRFY_ABHA_102	ABHA address login with Aadhaar OTP.	Hard	Uses ABDM's PHR web APIs (v3/phr/web/login/abha/...). The sandbox rejected abha-address on the profile login API, and the PHR documentation was not readable.
VRFY_ABHA_202	ABHA address login with mobile OTP.	Hard	The same PHR flow as above, so build both together.

2. Mandatory, built, but never tried with a real OTP (14)

This is testing, not coding, and it is where hidden bugs are most likely. Run each one in the browser against the ABDM sandbox.

Sheet row	What to try
CRT_ABHA_102	Consent shown in full and required before the OTP. Check the wording against ABDM's approved text.
CRT_ABHA_106	Resend OTP: available after 60 seconds, at most 2 times.
CRT_ABHA_109	Verify the communication mobile with an OTP when it is not the Aadhaar-linked one.
CRT_ABHA_112	ABHA address suggestions, custom address, rules shown beside the field (8 to 18 characters).
CRT_ABHA_113	ABHA number and address shown right after creation.
CRT_ABHA_114	ABHA card download (PNG or PDF) from the patient record.
VRFY_ABHA_101	Link by ABHA number, OTP to the Aadhaar-linked mobile.
VRFY_ABHA_201	Link by ABHA number, OTP to the ABHA's own mobile.
VRFY_ABHA_304, VRFY_ABHA_402	Wrong OTP shows a clear message (mobile and Aadhaar logins).
VRFY_ABHA_305, VRFY_ABHA_405	Resend limits on the mobile and Aadhaar logins.
VRFY_ABHA_403	An Aadhaar with no ABHA: correct message, and the create-ABHA option stays open.
VRFY_ABHA_404	Existing ABHA by Aadhaar: profile shown, linked, card downloadable.

3. Mandatory-adjacent gaps inside rows we marked as built (3)

Sheet row	Gap	Effort
VRFY_ABHA_101 and 201 (expected results 3 and 4)	Show the full profile after verification: name, date of birth, gender, mobile, photo, address, state, district. Name, date of birth and gender locked; the rest editable; saved against the patient. We show only name, date of birth, address and KYC.	Easy-Medium
CRT_ABHA_112	Show 'ABHA Address already exists' when a chosen address is taken (map ABDM's duplicate-address error).	Easy
CRT_ABHA_105	Match the sheet's example OTP wording ('We just sent an OTP on the Mobile Number *******XXXX linked with Aadhaar...').	Easy

4. Optional, easy to add (5)

Feature	Sheet row	Effort	Notes
Multilingual consent (Hindi first)	CRT_ABHA_103	Easy	Needs a reviewed translation. The Aadhaar rules say consent must be in the person's own language, so this is worth doing.
Offer to update the ABHA mobile (Yes/No) when it differs from the hospital record	VRFY_ABHA_101 and 201, step 5	Easy	Uses the profile OTP request and verify APIs.
Email update	PROF_ABHA_603	Easy	Same two APIs, with an OTP.
Mobile update	PROF_ABHA_601	Easy	Same two APIs, with an OTP.
Call ABDM's logout when staff sign out	Sheet row 149	Easy	One API call (v3/profile/account/request/logout).

5. Optional, medium effort (8)

Feature	Sheet row	Effort	Notes
Login with an ABHA password	Row 103	Medium	The password is RSA-encrypted like the OTP.
Set or change the ABHA password	Rows 110 to 114	Medium	Aadhaar OTP, mobile OTP or old-password routes.
Photo update	PROF_ABHA_602	Medium	Image upload with type and size checks.
Re-KYC	PROF_ABHA_604	Medium	Aadhaar OTP, then profile verify.
Delete or deactivate ABHA	PROF_ABHA_605	Medium	Destructive: needs a confirmation step and an audit entry.
Read a patient's ABHA QR (camera scan)	VRFY_ABHA_501 (QR row)	Medium	A scanner library, then parsing the QR payload.
Captcha on the mobile login	VRFY_ABHA_301	Medium	Marked 'preferred', not required.
Automated tests for NHA's functional scenarios	Outside the sheet	Medium	Timeout and resend, network failure mid-flow, cancel leaves no orphan, phone already registered.

6. Optional, hard (4)

Feature	Sheet rows	Effort	Why it is hard
Driving licence / PAN creation	CRT_ABHA_401 to 411	Hard	Mobile OTP, document number, licence photos (front and back), address, state, district and pincode. Also a staff step to manually check the documents and approve. Many new screens and sensitive document handling.
Aadhaar biometric creation and verification	CRT_ABHA_201 to 208, VRFY_ABHA_501 and 502	Hard	Needs a registered fingerprint device and UIDAI-compliant capture. ABDM lists no V3 API for the verification side.
Child ABHA	Row 136	Hard	Parent mapping plus the children APIs.
ABHA address creation through the PHR APIs	Row 140	Hard	Same unknown PHR API surface as the failing address-login rows.

7. Not pursuing

- Demo-auth / offline mode, CRT_ABHA_301 to 309: government entities only.
- Benefit programme APIs, rows 117 to 130: government benefit programmes (confirm with NHA).
- CRT_ABHA_209 and 210: only relevant if biometric creation is offered.

Suggested order

Step	Do this	Why
1	Test the 14 built rows (section 2) and fix what the real OTPs show.	Cheapest way to find real bugs before NHA's testers do.
2	Quick fixes in section 3.	Small, and they sit inside mandatory rows.
3	Deploy the backend, then build Scan and Share.	Mandatory, and it needs the public URL.
4	Get the PHR API documentation and build ABHA address login.	Two mandatory rows at once.
5	Pick from the easy and medium optional lists (multilingual consent first).	Strengthens the submission for little work.
6	Leave driving licence / PAN, biometrics and child ABHA unless NHA requires them for your category.	Large effort, optional for private integrators.

Also still ahead outside the sheet: NHA's functional-test run, the CERT-In security audit on a staging URL, a demo, and a second milestone (NHA needs two to leave the sandbox). Items marked N/A or not pursuing assume a private integrator; confirm with NHA.